

Concrete Mix Design Report – M30

Project: Commercial Building **Concrete Grade:** M30

Material / Parameter	Value	Unit	Remarks
Cement	380	kg/m³	Primary cementitious material
Fly Ash	80	kg/m³	Supplementary cementitious material
Water	170	L/m³	Batch water before moisture correction
Fine Aggregate / Sand	720	kg/m³	Natural/manufactured fine aggregate
10 mm Coarse Aggregate	420	kg/m³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m³	Nominal coarse aggregate fraction
Chemical Admixture	4.2	kg/m³	High-range water reducer
Water/Cementitious Ratio	0.370	-	Illustrative value
Target Slump	105	mm	At point of discharge
Target Mean Strength	38.25	MPa	Illustrative 28-day target

Design Notes

The M30 design uses a higher cementitious content and lower illustrative water-to-cementitious ratio than the M25 design. Slump is maintained using chemical admixture rather than simply increasing water.

Document status: Synthetic sample created for RMC RAG development and testing. Not an approved construction mix design. Actual proportions require laboratory validation, material test results, project specifications, moisture corrections, and applicable standards.